ADVANCED

Stage 6

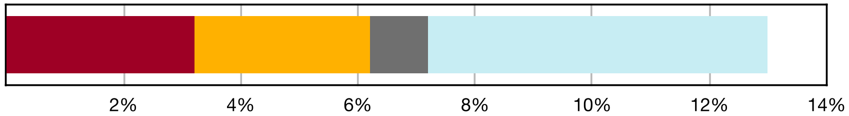
Financial Maths (Adv), M1 Financial Modelling (Adv)

Modelling Investments and Loans (Y12)

Teacher: SHS Maths

Exam Equivalent Time: 112.5 minutes *(based on allocation of 1.5 minutes per mark)*

M1 Financial Modelling



- Investments and Loans
- Arithmetics Series
- Geometric Series
- Financial Applications of Series

***SmarterMaths analytics based on the average contribution to new syllabus Advanced Maths exams since 2020.**

HISTORICAL CONTRIBUTION

- *M1 Financial Modelling* has contributed a substantial 13.0% to new syllabus Advanced exams since it was introduced in 2020.
- The topic is split into four sub-topics for the purposes of analysis: *1-Investments and Loans (3.2%), 2-Arithmetic Series (3.0%), 3-Geometric Series (1.0%) and 4-Financial Applications of Series (5.8%).*
- This analysis looks at *Investments and Loans.*

HSC ANALYSIS - What to expect and common pitfalls

- *Investments and Loans (3.2%)* has attracted a substantial 4-5 mark longer answer question (note all were common to both Std2/Adv exams) each year in the period 2022-24.
- The *Future Value of an Annuity Table* is the most common question type, examined each year between 2021-23. Important revision examples include *2022 Adv 21, 2019 Std2 42* and *M1 EQ-Bank 1*.
- *Present value interest factors* is another important question type, contributing a high difficulty challenge in *2024 Adv 26* (see also *2015 Std2 30c)*.
- Applying the compound interest formula $A = P(1+r)^n$ is a core skill and important revision.
- *Compounded Value of $1* tables has proven challenging in past Std2 exams. Needs to be specifically revised although we note it is yet to appear in an Advanced exam.

Questions

1. Financial Maths, STD2 F4 2009 HSC 6 MC

A house was purchased in 1984 for $35 000. Assume that the value of the house has increased by 3% per annum since then.

Which expression gives the value of the house in 2009?

A. $35\ 000(1+0.03)^{25}$

B. $35\ 000(1+3)^{25}$

C. $35\ 000 \times 25 \times 0.03$

D. $35\ 000 \times 25 \times 3$

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2. Financial Maths, STD2 F4 2015 HSC 17 MC

What amount must be invested now at 4% per annum, compounded quarterly, so that in five years it will have grown to $60 000?

A. $8919

B. $11 156

C. $49 173

D. $49 316

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3. Financial Maths, STD2 F4 2016 HSC 8 MC

The table shows the future value of an investment of $1000, compounding yearly, at varying interest rates for different periods of time.

Future values of an investment of $1000

Number of years	*Interest rate per annum*				
	1%	2%	3%	4%	5%
1	1010.00	1020.00	1030.00	1040.00	1050.00
2	1020.10	1040.40	1060.90	1081.60	1102.50
3	1030.30	1061.21	1092.73	1124.86	1157.63
4	1040.60	1082.43	1125.51	1169.86	1215.51
5	1051.01	1104.08	1159.27	1216.65	1276.28

Based on the information provided, what is the future value of an investment of $2500 over 3 years at 4% pa?

A. $1124.86

B. $2812.15

C. $3624.86

D. $5312.15

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4. Financial Maths, STD2 F4 2017 HSC 10 MC

A single amount of \$10 000 is invested for 4 years, earning interest at the rate of 3% per annum, compounded monthly.

Which expression will give the future value of the investment?

A. $10\ 000 \times (1+0.03)^4$

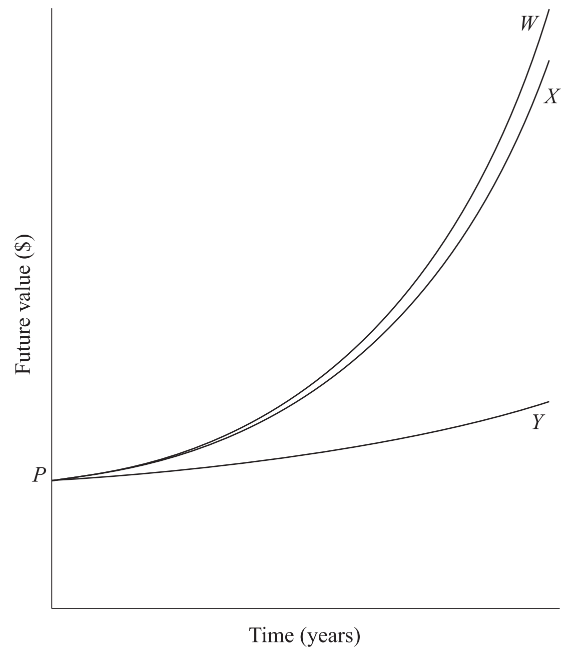
B. $10\ 000 \times (1+0.03)^{48}$

C. $10\ 000 \times \left(1+\frac{0.03}{12}\right)^4$

D. $10\ 000 \times \left(1+\frac{0.03}{12}\right)^{48}$

5. Financial Maths, STD2 F4 2019 HSC 13 MC

The graph show the future values over time of $\$P$, invested at three different rates of compound interest.



Which of the following correctly identifies each graph?

A.

W	5% pa, compounding annually
X	10% pa, compounding annually
Y	10% pa, compounding quarterly

B.

W	5% pa, compounding annually
X	10% pa, compounding quarterly
Y	10% pa, compounding annually

C.

W	10% pa, compounding quarterly
X	10% pa, compounding annually
Y	5% pa, compounding annually

D.

W	10% pa, compounding annually
X	10% pa, compounding quarterly
Y	5% pa, compounding annually

6. Financial Maths, STD2 F4 2019 HSC 3 MC

Chris opens a bank account and deposits $1000 into it. Interest is paid at 3.5% per annum, compounding annually.

Assuming no further deposits or withdrawals are made, what will be the balance in the account at the end of two years?

A. $1070.00

B. $1071.23

C. $1822.50

D. $2070.00

7. Financial Maths, STD2 F2 2021 HSC 5 MC

Peter currently earns $21.50 per hour. His hourly wage will increase by 2.1% compounded each year for the next four years.

What will his hourly wage be after four years?

A. $21.50(1.21)^4$

B. $21.50(1.021)^4$

C. $21.50 + 21.50 \times 0.21 \times 4$

D. $21.50 + 21.50 \times 0.021 \times 4$

8. Financial Maths, STD2 F4 2012 HSC 9 MC

Tracy invests some money for 2 years at 4% per annum, compounded quarterly.

Compounded values of $1

Period	Interest rate per period				
	1%	*2%*	*3%*	*4%*	*5%*
1	1.010	1.020	1.030	1.040	1.050
2	1.020	1.040	1.061	1.082	1.103
3	1.030	1.061	1.093	1.125	1.158
4	1.041	1.082	1.126	1.170	1.216
5	1.051	1.104	1.159	1.217	1.276
6	1.062	1.126	1.194	1.265	1.340
7	1.072	1.149	1.230	1.316	1.407
8	1.083	1.172	1.267	1.369	1.477

Which figure from the table should Tracy use to calculate the value of her investment at the end of 2 years?

A. 1.020

B. 1.082

C. 1.083

D. 1.369

9. Financial Maths, STD2 F5 2014 HSC 21 MC

A table of future value interest factors is shown.

Table of future value interest factors

Period	*Interest rate per period*				
	1%	*2%*	*3%*	*4%*	*5%*
1	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500
3	3.0301	3.0604	3.0909	3.1216	3.1525
4	4.0604	4.1216	4.1836	4.2465	4.3101

A certain annuity involves making equal contributions of $25 000 into an account every 6 months for 2 years at an interest rate of 4% per annum.

Based on the information provided, what is the future value of this annuity?

A. $\$50\ 500$

B. $\$51\ 000$

C. $\$103\ 040$

D. $\$106\ 162$

10. Financial Maths, STD2 F4 2018 HSC 19 MC

The table shows the compounded values of $1 at different interest rates over different periods.

Compounded values of $1

Number of periods	*Interest rate per period*				
	1%	2%	3%	4%	5%
2	1.0201	1.0404	1.0609	1.0816	1.1025
4	1.0406	1.0824	1.1255	1.1699	1.2155
6	1.0615	1.1262	1.1941	1.2653	1.3401
8	1.0829	1.1717	1.2668	1.3686	1.4775
10	1.1046	1.2190	1.3439	1.4802	1.6289
12	1.1268	1.2682	1.4258	1.6010	1.7959

Amy hopes to have $21 000 in 2 years to buy a car. She opens an account today which pays interest of 4% pa, compounded quarterly.

Using the table, which expression calculates the minimum single sum that Amy needs to invest today to ensure she reaches her savings goal?

A. 21 000 × 1.0816

B. 21 000 ÷ 1.0816

C. 21 000 × 1.0829

D. 21 000 ÷ 1.0829

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11. Financial Maths, STD2 F5 2013 23 MC

Zina opened an account to save for a new car. Six months after opening the account, she made first deposit of $1200 and continued depositing $1200 at the end of each six month period. Interest was paid at 3% per annum, compounded half-yearly.

How much was in Zina's account two years after first opening it?

A. $4909.08

B. $4982.72

C. $5018.16

D. $5094.55

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12. Financial Maths, 2ADV M1 2023 HSC 15

A table of future value interest factors for an annuity of $1 is shown.

Period \ *Rate*	*1.5%*	*3%*	*4.5%*	*6%*
5	5.152	5.309	5.471	5.637
10	10.703	11.464	12.288	13.181
20	23.124	26.870	31.371	36.786
40	54.268	75.401	107.030	154.762

a. Micky wants to save $450 000 over the next 10 years.

If the interest rate is 6% per annum compounding annually, how much should Micky contribute each year? Give your answer to the nearest dollar. ***(2 marks)***

b. Instead, Micky decides to contribute $8535 every three months for 10 years to an annuity paying 6% per annum, compounding quarterly.

How much will Micky have at the end of 10 years? ***(3 marks)***

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13. Financial Maths, STD2 F5 2018 HSC 26c

Ali made monthly deposits of $100 into an annuity for 5 years.

Calculate the total amount Ali deposited into the annuity over this period. ***(1 mark)***

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14. Financial Maths, STD2 F5 SM-Bank 2

The table below shows the present value of an annuity with a contribution of $1.

Present Value of Annuity of $1				
Period	1%	2%	4%	6%
1	0.9901	0.9804	0.9615	0.9434
2	1.9704	1.9416	1.8861	1.8334
3	2.9410	2.8839	2.7751	2.6730
4	3.9020	3.8077	3.6299	3.4651

i. Fiona pays $3000 into an annuity at the end of each year for 4 years at 2% p.a., compounded annually. What is the present value of her annuity? ***(1 mark)***

ii. If John pays $6000 into an annuity at the end of each year for 2 years at 4% p.a., compounded annually, is he better off than Fiona? Use calculations to justify your answer. ***(2 marks)***

15. Financial Maths, STD2 F5 SM-Bank 4

Dominique wants to save $15 000 to use as spending money when she travels overseas in 2 years' time.

If she invests $3500 at the end of every 6 months into an account earning 4% p.a., compounded half-yearly, will she have enough?

Use the table below to justify your answer. ***(2 marks)***

Periods (n)	Future Value of an Annuity of $1				
	Interest Rate				
	2%	3%	4%	5%	6%
2	2.020	2.030	2.040	2.050	2.060
4	4.122	4.184	4.247	4.310	4.375
6	6.308	6.468	6.633	6.802	6.975
8	8.583	8.892	9.214	9.549	9.898
10	10.950	11.464	12.006	12.578	13.181

16. Financial Maths, 2ADV M1 2021 HSC 25

A table of future value interest factors for an annuity of $1 is shown.

Table of future value interest factors

Number of periods	*Interest rate per period*				
	0.25%	0.5%	0.75%	1%	1.25%
2	2.0025	2.0050	2.0075	2.0100	2.0125
4	4.0150	4.0301	4.0452	4.0604	4.0756
6	6.0376	6.0755	6.1136	6.1520	6.1907
8	8.0704	8.1414	8.2132	8.2857	8.3589
10	10.1133	10.2280	10.3443	10.4622	10.5817

Simone deposits $1000 into a savings account at the end of each year for 8 years. The interest rate for these 8 years is 0.75% per annum, compounded annually.

After the 8th deposit, Simone stops making deposits but leaves the money in the savings account. The money in her savings account then earns interest at 1.25% per annum, compounded annually, for a further two years.

Find the amount of money in Simone's savings account at the end of ten years. ***(3 marks)***

17. Financial Maths, 2ADV M1 EQ-Bank 1

Ralph opens an annuity account and makes a contribution of $12 000 at the end of each year for 9 years.

For the first 8 years, the interest rate is 4% per annum, compounded annually.

For the 9th year, the interest rate decreases to 3% per annum, compounded annually.

Periods (n)	Future Value of an Annuity of $1				
	Interest Rate				
	2%	3%	4%	5%	6%
2	2.020	2.030	2.040	2.050	2.060
4	4.122	4.184	4.247	4.310	4.375
6	6.308	6.468	6.633	6.802	6.975
8	8.583	8.892	9.214	9.549	9.898
10	10.950	11.464	12.006	12.578	13.181

Use the Future Value of an Annuity table to calculate the amount in the account immediately after the 9th contribution is made. ***(3 marks)***

18. Financial Maths, STD2 F4 2008 HSC 24c

Heidi's funds in a superannuation scheme have a future value of $740 000 in 20 years time. The interest rate is 4% per annum and earnings are calculated six-monthly.

What single amount could be invested now to produce the same result over the same period of time at the same interest rate? ***(3 marks)***

19. Financial Maths, STD2 F4 2015 HSC 26d

A family currently pays $320 for some groceries.

Assuming a constant annual inflation rate of 2.9%, calculate how much would be paid for the same groceries in 5 years' time. ***(2 marks)***

20. Financial Maths, 2ADV M1 2022 HSC 21

Eli is choosing between two investment options.

Option 1: Depositing a single amount of $40 000 today, earning interest of 1.2% per annum, compounded monthly.

Option 2: Depositing $1000 at the end of each quarter, earning interest of 2.4% per annum, compounded quarterly.

A table of future value interest factors for an annuity of $1 is shown.

N \ r	*Interest rate per period as a decimal*					
	0.002	0.006	0.020	0.024	0.060	0.240
10	10.09048	10.27437	10.94972	11.15211	13.18079	31.64344
20	20.38460	21.18211	24.29737	25.28909	36.78559	303.60062
30	30.88646	32.76227	40.56808	43.20983	79.05819	2640.91639
40	41.60026	45.05630	60.40198	65.92708	154.76197	22 728.80260

a. What is the value of Eli's investment after 10 years using Option 1 ? ***(2 marks)***

b. What is the difference between the future values after 10 years using Option 1 and Option 2? ***(2 marks)***

21. Financial Maths, STD2 F5 2009 HSC 27a

The table shows the future value of a $1 annuity at different interest rates over different numbers of time periods.

Future values of a $1 annuity

Time Period	*Interest rate*				
	1%	*2%*	*3%*	*4%*	*5%*
1	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500
3	3.0301	3.0604	3.0909	3.1216	3.1525
4	4.0604	4.1216	4.1836	4.2465	4.3101
5	5.1010	5.2040	5.3091	5.4163	5.5256
6	6.1520	6.3081	6.4684	6.6330	6.8019
7	7.2135	7.4343	7.6625	7.8983	8.1420
8	8.2857	8.5830	8.8923	9.2142	9.5491

i. What would be the future value of a $5000 per year annuity at 3% per annum for 6 years, with interest compounding yearly? ***(1 mark)***

ii. What is the value of an annuity that would provide a future value of $407100 after 7 years at 5% per annum compound interest? ***(1 mark)***

iii. An annuity of $1000 per quarter is invested at 4% per annum, compounded quarterly for 2 years. What will be the amount of interest earned? ***(3 marks)***

22. Financial Maths, STD2 F5 2005 HSC 26b

Rod is saving for a holiday. He deposits $3600 into an account at the end of every year for four years. The account pays 5% per annum interest, compounding annually.

The table shows future values of an annuity of $1.

Future values of an annuity of $1

End of year	***Interest rate***				
	1%	**2%**	**3%**	**4%**	**5%**
1	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500
3	3.0301	3.0604	3.0909	3.1216	3.1525
4	4.0604	4.1216	4.1836	4.2465	4.3101
5	5.1010	5.2040	5.3091	5.4163	5.5256
6	6.1520	6.3081	6.4684	6.6330	6.8019
7	7.2135	7.4343	7.6625	7.8983	8.1420
8	8.2857	8.5830	8.8923	9.2142	9.5491

i. Use the table to find the value of Rod's investment at the end of four years. ***(2 marks)***

ii. How much interest does Rod earn on his investment over the four years? ***(2 marks)***

23. Financial Maths, STD2 F5 2017 HSC 27c

A table of future value interest factors for an annuity of $1 is shown.

Table of future value interest factors					
Period	*Interest rate per period*				
	1%	2%	3%	4%	5%
3	3.0301	3.0604	3.0909	3.1216	3.1525
4	4.0604	4.1216	4.1836	4.2465	4.3101
5	5.1010	5.2040	5.3091	5.4163	5.5256
6	6.1520	6.3081	6.4684	6.6330	6.8019

An annuity involves contributions of $12 000 per annum for 5 years. The interest rate is 4% per annum, compounded annually.

i. Calculate the future value of this annuity. ***(1 mark)***

ii. Calculate the interest earned on this annuity. ***(1 mark)***

24. Financial Maths, STD2 F5 SM-Bank 3

Camilla buys a car for $21 000 and repays it over 4 years through equal monthly instalments.

She pays a 10% deposit and interest is charged at 9% p.a. on the reducing balance loan.

Using the Table of present value interest factors below, where r represents the monthly interest and N represents the number of repayments

Table of Present Value Interest Factors						
r	**0.0060**	**0.0065**	**0.0070**	**0.0075**	**0.0080**	**0.0085**
N						
45	39.33406	38.90738	38.48712	38.07318	37.66545	37.26383
46	40.09350	39.64965	39.21263	38.78231	38.35859	37.94133
47	40.84841	40.38714	39.93310	39.48617	39.04622	38.61311
48	41.59882	41.11986	40.64856	40.18478	39.72839	39.27924
49	42.34475	41.84785	41.35905	40.87820	40.40515	39.93975
50	43.08623	42.57113	42.06459	41.56645	41.07653	40.59470

i. Calculate the monthly repayment, $\$P$, that Camilla must pay to complete the loan after 4 years (to the nearest $). ***(3 marks)***

ii. Calculate the total interest paid over the life of the loan. ***(1 mark)***

25. Financial Maths, STD2 F5 2011 HSC 27d

Josephine invests $50 000 for 15 years, at an interest rate of 6% per annum, compounded annually.

Emma invests $500 at the end of each month for 15 years, at an interest rate of 6% per annum, compounded monthly.

Financial gain is defined as the difference between the final value of an investment and the total contributions.

Who will have the better financial gain after 15 years? Using the Table below* and appropriate formulas, justify your answer with suitable calculations. ***(4 marks)***

Future value of an annuity of $1				
Period	0.50%	1%	2%	3%
120	163.8793	230.0387	488.2582	1123.6996
180	290.8187	499.5802	1716.0416	6783.4453
240	462.0409	989.2554	5744.4368	40128.4209

26. Financial Maths, STD2 F4 2014 HSC 30a

Chandra and Sascha plan to have $20 000 in an investment account in 15 years time for their grandchild's university fees.

The interest rate for the investment account will be fixed at 3% per annum compounded monthly.

Calculate the amount that they will need to deposit into the account now in order to achieve their plan. ***(3 marks)***

27. Financial Maths, STD2 F4 2011 HSC 23c

An amount of $5000 is invested at 10% per annum, compounded six-monthly.

Compounded values of $1

Period	**Interest rate per period**				
	1%	*5%*	*10%*	*15%*	*20%*
1	1.010	1.050	1.100	1.150	1.200
2	1.020	1.103	1.210	1.323	1.440
3	1.030	1.158	1.331	1.521	1.728
4	1.041	1.216	1.464	1.750	2.074
5	1.051	1.276	1.611	2.011	2.488
6	1.062	1.340	1.772	2.313	2.986

Use the table to find the value of this investment at the end of three years. ***(2 marks)***

28. Financial Maths, STD2 F4 2013 HSC 26e

Kimberley has invested $3500.

Interest is compounded half-yearly at a rate of 2% per half-year.

Compounded values of $1

Period	*Interest rate per period*					
	1%	2%	3%	4%	5%	6%
1	1.010	1.02	1.03	1.04	1.05	1.06
2	1.020	1.040	1.061	1.082	1.103	1.124
3	1.030	1.061	1.093	1.125	1.158	1.191
4	1.041	1.082	1.126	1.170	1.216	1.262
5	1.051	1.104	1.159	1.217	1.276	1.338
6	1.062	1.126	1.194	1.265	1.340	1.419
7	1.072	1.149	1.230	1.316	1.407	1.504
8	1.083	1.172	1.267	1.369	1.477	1.594

Use the table to calculate the value of her investment at the end of 4 years. ***(2 marks)***

29. Financial Maths, 2ADV M1 2024 HSC 26

Twenty-five years ago, Phoenix deposited a single sum of money into a new bank account, earning 2.4% interest per annum compounding monthly.

Present value interest factors for an annuity of $1 for various interest rates (r) and numbers of periods (n) are given in the table.

Period (n) \ *Rate (r)*	*0.001*	*0.002*	*0.003*	*0.004*
60	58.207	56.487	54.835	53.249
120	113.026	106.592	100.649	95.156
180	164.655	151.036	138.927	128.137
240	213.278	190.460	170.908	154.093
300	259.071	225.430	197.627	174.521

Phoenix made the following withdrawals from this account.

- $2000 at the end of each month for the first 15 years, starting at the end of the first month.
- $1200 at the end of each month for the next 10 years, starting at the end of the 181st month after the account was opened.

Calculate the minimum sum that Phoenix could have deposited in order to make these withdrawals. ***(4 marks)***

30. Financial Maths, STD2 F5 2015 HSC 30c

The table gives the present value interest factors for an annuity of $1 per period, for various interest rates (r) and numbers of periods (N).

Table of present value interest factors

N \ r	Interest rate per period (as a decimal)				
	0.0075	0.0080	0.0085	0.0090	0.0095
70	54.30462	53.43960	52.59397	51.76724	50.95891
71	54.89293	54.00754	53.14226	52.29657	51.46995
72	55.47685	54.57097	53.68593	52.82118	51.97618
73	56.05643	55.12993	54.22502	53.34111	52.47764
74	56.63169	55.68446	54.75957	53.85641	52.97438

i. Oscar plans to invest $200 each month for 74 months. His investment will earn interest at the rate of 0.0080 (as a decimal) per month.
   Use the information in the table to calculate the present value of this annuity. ***(1 mark)***

ii. Lucy is using the same table to calculate the loan repayment for her car loan. Her loan is $\$21\ 500$ and will be repaid in equal monthly repayments over 6 years. The interest rate on her loan is 10.8% per annum.
   Calculate the amount of each monthly repayment, correct to the nearest dollar. ***(2 marks)***

31. Financial Maths, STD2 F5 2016 HSC 28d

The table gives the contribution per period for an annuity with a future value of $1 at different interest rates and different periods of time.

Contribution per period for an annuity with a future value of $1

Number of periods	*Interest rate (% per period)*					
	0.25%	0.5%	0.75%	1%	1.25%	1.5%
6	0.1656	0.1646	0.1636	0.1625	0.1615	0.1605
12	0.0822	0.0811	0.0800	0.0788	0.0778	0.0767
18	0.0544	0.0532	0.0521	0.0510	0.0499	0.0488
24	0.0405	0.0393	0.0382	0.0371	0.0360	0.0349
30	0.0321	0.0310	0.0298	0.0287	0.0277	0.0266
36	0.0266	0.0254	0.0243	0.0232	0.0222	0.0212

Margaret needs to save $75 000 over 6 years for a deposit on a new apartment. She makes regular quarterly contributions into an investment account which pays interest at 3% pa.

How much will Margaret need to contribute each quarter to reach her savings goal? ***(2 marks)***

32. Financial Maths, STD2 F5 2019 HSC 42

The table shows the future values of an annuity of $1 for different interest rates for 4, 5 and 6 years. The contributions are made at the end of each year.

Future value of an annuity of $1

Years	*Interest rate per annum*			
	1%	*2%*	*3%*	*4%*
4	4.060	4.122	4.184	4.246
5	5.101	5.204	5.309	5.416
6	6.152	6.308	6.468	6.633

An annuity account is opened and contributions of $2000 are made at the end of each year for 7 years.

For the first 6 years, the interest rate is 4% per annum, compounding annually.

For the 7th year, the interest rate increases to 5% per annum, compounding annually.

Calculate the amount in the account immediately after the 7th contribution is made. ***(3 marks)***

Worked Solutions

1. Financial Maths, STD2 F4 2009 HSC 6 MC

$r = 3\% = 0.03$
$n = 25$ years
Using $FV = PV(1+r)^n$
$\therefore$ Value in 2009 $= 35\ 000(1+0.03)^{25}$
$\Rightarrow A$

2. Financial Maths, STD2 F4 2015 HSC 17 MC

Using $FV = PV(1+r)^n$

$r = \dfrac{4\%}{4} = 1\% = 0.01$ per quarter
$n = 5 \times 4 = 20$ quarters

$60\ 000 = PV(1+0.01)^{20}$

$\therefore PV = \dfrac{60\ 000}{1.01^{20}}$

$= \$49\ 172.66\ldots$

$\Rightarrow C$

3. Financial Maths, STD2 F4 2016 HSC 8 MC

Table factor $= 1124.86$
$\therefore FV = 2.5 \times 1124.86$
$= \$2812.15$

$\Rightarrow B$

4. Financial Maths, STD2 F4 2017 HSC 10 MC

Compounding rate $= \dfrac{3}{100} \div 12 = \dfrac{0.03}{12}$

Compounding periods $= 4 \times 12 = 48$

$\therefore$ FV $= 10\ 000 \times \left(1 + \dfrac{0.03}{12}\right)^{48}$

$\Rightarrow D$

5. Financial Maths, STD2 F4 2019 HSC 13 MC

Values increase quicker

- higher compounding interest rate
- same rate but more frequent compounding period

$\therefore W = 10\%$ quarterly

$X = 10\%$ annually

$Y = 5\%$ annually

$\Rightarrow C$

6. Financial Maths, STD2 F4 2019 HSC 3 MC

$$FV = PV(1+r)^n$$
$$= 1000(1+0.035)^2$$
$$= \$1071.23$$

$\Rightarrow B$

7. Financial Maths, STD2 F2 2021 HSC 5 MC

Wage after 1 year $= 21.50 \times 1.021$

Wage after 2 years $= 21.50 \times 1.021 \times 1.021 = 21.50(1.021)^2$

$\vdots$

Wage after 4 years $= 21.50(1.021)^4$

$\Rightarrow B$

8. Financial Maths, STD2 F4 2012 HSC 9 MC

4% annual

$\Rightarrow \dfrac{4\%}{4} = 1\%$ compounded each quarter

$\Rightarrow n = 8$ (8 quarters in 2 years)

$\therefore$ Factor $= 1.083$ (from table)

$\Rightarrow C$

♦♦♦ *Mean mark 24%. The lowest MC mean mark in the 2012 exam.*
COMMENT: A process to follow: 1-convert the annual rate to the rate per compounding period. 2-calculate the number of compounding periods.

9. Financial Maths, STD2 F5 2014 HSC 21 MC

4 contributions of \$25 000 made.

Annuity period $= 6$ months

Rate (per annuity period) $= \dfrac{4\%}{2} = 2\%$

Periods $= 4$ (4 x 6 months $= 2$ years)

Table value $= 4.1216$

$\therefore$ Annuity Value $= 4.1216 \times 25\ 000 = \$103\ 040$

$\Rightarrow C$

♦ *Mean mark 43%*

10. Financial Maths, STD2 F4 2018 HSC 19 MC

4% annual

$\Rightarrow \dfrac{4\ \%}{4} = 1\%$ compounded quarterly

$\Rightarrow n = 8$

$\Rightarrow$ Factor $= 1.0829$

$\therefore$ Minimum sum $= 21\ 000 \div 1.0829$

$\Rightarrow$ D

♦♦ *Mean mark 33%.*

11. Financial Maths, STD2 F5 2013 23 MC

Interest: 3% p.a $\Rightarrow$ 1.5% per 6 months

After 2 years,

Value of 1st deposit $= 1200(1.015)^3 = 1254.81$

Value of 2nd deposit $= 1200(1.015)^2 = 1236.27$

Value of 3rd deposit $= 1200(1.015) = 1218$

Value of 4th deposit $= 1200$

$\therefore$ Amount in account after 2 years

$= 1254.81 + 1236.27 + 1218 + 1200$

$= \$4909.08$

$\Rightarrow A$

♦ *Mean mark 41%.*

12. Financial Maths, 2ADV M1 2023 HSC 15

a. Applicable interest rate $= 6\,\%$

Compounding periods $= 10 \times 1 = 10$

$\Rightarrow$ Factor $= 13.181$

$$\therefore \text{Contribution (annual)} = \frac{450\ 000}{13.181} = \$34\ 140$$

b. Applicable interest rate $= \dfrac{6\,\%}{4} = 1.5\,\%$ per quarter

Compounding periods $= 10 \times 4 = 40$

$\Rightarrow$ Factor $= 54.268$

$$\text{Total (after 10 years)} = 8535 \times 54.268 = \$463\ 177.38$$

13. Financial Maths, STD2 F5 2018 HSC 26c

$$\text{Total deposited} = 5 \times 12 \times 100 = \$6000$$

14. Financial Maths, STD2 F5 SM-Bank 2

i. Table factor when $n = 4,\ r = 2\% \Rightarrow 3.8077$

$$\therefore PVA\ \text{(Fiona)} = 3000 \times 3.8077 = \$11\ 423.10$$

ii. Table factor when $n = 2, r = 4\%$

$\Rightarrow 1.8861$

$$\therefore PVA\ \text{(John)} = 6000 \times 1.8861 = \$11\ 316.60$$

$\therefore$ Fiona will be better off because her PVA is higher.

15. Financial Maths, STD2 F5 SM-Bank 4

Interest rate (6 monthly) $= 4\% \div 2 = 2\%$

$n = 4$ (6 month periods in 2 years)

$\Rightarrow$ FVA factor $= 4.122$ (from Table)

$$\text{FVA} = 3500 \times 4.122 = \$14\ 427$$

$\therefore$ Dominique will not have saved enough to reach $15 000.

16. Financial Maths, 2ADV M1 2021 HSC 25

Mean mark 52%.

In 1st 8 years:

Future value factor $= 8.2132$

$$\text{Value of annuity} = 8.2132 \times 1000 = \$8213.20$$

After 10 years:

$$\text{Value of investment} = 8213.2 \times (1.0125)^2 = \$\$8419.81$$

17. Financial Maths, 2ADV M1 EQ-Bank 1

After 8 years:

$$\text{Total} = 9.214 \times \$12\ 000 = \$110\ 568$$

After 9 years (9th contribution made year end):

$$\text{Total} = 110\ 568 \times 1.03 + 12\ 000 = \$125\ 885.04$$

18. Financial Maths, STD2 F4 2008 HSC 24c

$$FV = PV(1+r)^n$$

$$740\ 000 = PV\left(1 + \frac{2}{100}\right)^{40}$$

$$\therefore PV = \frac{740\ 000}{(1.02)^{40}} = 335\ 138.907\ldots = \$335\ 138.91$$

19. Financial Maths, STD2 F4 2015 HSC 26d

$$\begin{aligned} FV &= PV(1+r)^n \\ &= 320(1.029)^5 \\ &= \$369.1703\ldots \\ &= \$369.17 \quad \text{(nearest cent)} \end{aligned}$$

20. Financial Maths, 2ADV M1 2022 HSC 21

a. Monthly r/i $= \dfrac{1.2}{12} = 0.1\% \quad \Rightarrow \quad r = 0.001$

Compounding periods $(n) = 12 \times 10 = 120$

$$\begin{aligned} FV &= PV(1+r)^n \\ &= 40\ 000(1+0.001)^{120} \\ &= \$45\ 097.17 \end{aligned}$$

b. Quarterly r/i $= \dfrac{2.4}{4} = 0.6\% \quad \Rightarrow \quad r = 0.006$

Compounding periods $(N) = 4 \times 10 = 40$

Annuity factor (from table) $= 45.05630$

$$\begin{aligned} FV &= 1000 \times 45.05630 \\ &= 45\ 056.30 \end{aligned}$$

$$\begin{aligned} \text{Difference} &= 45\ 097.17 - 45\ 056.30 \\ &= \$40.87 \end{aligned}$$

21. Financial Maths, STD2 F5 2009 HSC 27a

i. Table factor when $n = 6, \quad r = 3\% \Rightarrow 6.4684$

$$\begin{aligned} \therefore\ FV &= 5000 \times 6.4684 \\ &= \$32\ 342 \end{aligned}$$

ii. Table factor when $n = 7, \quad r = 5\%$

$\Rightarrow 8.1420$

Let $A =$ annuity

$$\begin{aligned} FV &= A \times 8.1420 \\ A &= \frac{FV}{8.1420} \\ &= \frac{407\ 100}{8.1420} \\ &= \$50\ 000 \end{aligned}$$

♦ *Mean mark 45%*
MARKER'S COMMENT: A common error was to multiply $407 100 by 8.1420 rather than divide.

iii. $n = 8 \quad$ (8 quarters in 2 years)

$r = \dfrac{4\%}{4} = 1\%$ per quarter

$\therefore$ Table factor $\Rightarrow 8.2857$

$$\begin{aligned} FV &= 1000 \times 8.2857 \\ &= 8285.70 \end{aligned}$$

$$\begin{aligned} \text{Interest} &= FV(\text{annuity}) - \text{Principal} \\ &= 8285.70 - (8 \times 1000) \\ &= 285.70 \end{aligned}$$

$\therefore$ Interest earned is \$285.70

♦♦ *Mean mark 31%*
MARKER'S COMMENT: When questions asked for the interest paid on annuities, remember to subtract the total principal amounts contributed.

22. Financial Maths, STD2 F5 2005 HSC 26b

i. Using the table, $r = 5\%$ and $n = 4$

Annuity factor $= 4.3101$

$\therefore$ Value of investment

$$\begin{aligned} &= 3600 \times 4.3101 \\ &= \$15\ 516.36 \end{aligned}$$

ii.
$$\begin{aligned} \text{Interest} &= \text{Value} - \text{Contributions} \\ &= 15\ 516.36 - (4 \times 3600) \\ &= \$1116.36 \end{aligned}$$

23. Financial Maths, STD2 F5 2017 HSC 27c

i. FV factor $= 5.4163$

$$\therefore \text{ FV of Annuity } = 12\ 000 \times 5.4163$$
$$= \$64\ 995.60$$

♦♦ Mean mark part (ii) 22%.
COMMENT: A very poorly answered question dealing with a core concept in this area.

ii. Interest earned $=$ FV $-$ total repayments

$$= 64\ 995.60 - (5 \times 12\ 000)$$
$$= \$4995.60$$

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24. Financial Maths, STD2 F5 SM-Bank 3

i. Deposit $= 10\% \times 21\ 000 = 2100$

$$\text{Loan Value} = 21\ 000 - 2100$$
$$= 18\ 900$$

$$\text{Monthly interest rate} = \frac{9\%}{12} = 0.0075$$
$\#$ Repayments $= 4 \times 12 = 48$
$\Rightarrow$ PVA Factor $= 40.18478$ (from Table)

$$\text{Monthly repayment } (\$P) = \frac{18\ 900}{40.18478}$$
$$= 470.32\ldots$$
$$= 470 \text{ (nearest \$)}$$

$\therefore$ Camilla must repay \$470 per month.

ii. Total Repayments
$$= 48 \times 470$$
$$= \$22\ 560$$

$\therefore$ Interest paid over loan
$$= 22\ 560 - 18\ 900$$
$$= \$3660$$

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25. Financial Maths, STD2 F5 2011 HSC 27d

♦ Mean mark 42%
COMMENT: Note that compound interest vs annuity comparisons are commonly tested.

Josephine

$$\text{Investment} = 50\ 000(1 + 0.06)^{15}$$
$$= 50\ 000(1.06)^{15}$$
$$= \$119\ 827.91$$

$$\text{Financial gain} = 119\ 827.91 - 50\ 000$$
$$= \$69\ 827.91$$

Emma
Monthly interest rate $= 6\% \div 12 = 0.5\%$
$\#$ Monthly Payments $= 12 \times 15 = 180$
$\Rightarrow$ Annuity Factor $= 290.8187$ (from Table)

$$\text{Investment} = 500 \times 290.8187$$
$$= \$145\ 409.35$$

$$\text{Financial gain} = 145\ 409.35 - \text{total contributions}$$
$$= 145\ 409.35 - (500 \times 12 \times 15)$$
$$= 145\ 409.35 - 90\ 000$$
$$= \$55\ 409.35$$

$\therefore$ Josephine will have the better financial gain.

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26. Financial Maths, STD2 F4 2014 HSC 30a

♦ Mean mark 49%

$FV = \$20\ 000,\quad n = 15 \times 12 = 180,$

$$r = \frac{0.03}{12} = 0.0025$$

$$FV = PV(1+r)^n$$
$$20\ 000 = PV(1 + 0.0025)^{180}$$
$$PV = \frac{20\ 000}{(1.0025)^{180}}$$
$$= 12\ 759.73\ldots$$

$\therefore$ They need to deposit \$12 760 (nearest \$)

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27. Financial Maths, STD2 F4 2011 HSC 23c

Interest rate $= 10\%$ pa $= 5\%$ per 6 months

Period $= 6 \quad$ (6 x 6 months in 3 years)

$\Rightarrow$ Table value $= 1.340$

$\therefore$ Value of investment $= 5000 \times 1.34$

$= \$6700$

♦♦ Mean mark 28%
MARKER'S COMMENT: Remember that the number of periods is the number of "compounding periods" and when asked to use the table, *use the table!*

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28. Financial Maths, STD2 F4 2013 HSC 26e

$r =$ 2% per half-year

$n = 8 \quad$ (8 half-years in 4 years)

$\Rightarrow$ Table Factor $= 1.172$

Investment $= 3500 \times 1.172$

$= \$4102$

$\therefore$ After 4 years, investment value is \$4102

♦ Mean mark 44%
COMMENT: Structure your answer: 1-Find the interest rate per compounding period (same in this case). 2-Find the number of *compounding periods*.

---

29. Financial Maths, 2ADV M1 2024 HSC 26

1st Annuity

Find PVA for \$2000 paid monthly for 1st 15 years:

$$r = \frac{2.4}{12} = 0.2\% = 0.002$$

Total payments (to Phoenix) $= 15 \times 12 = 180$

PVA factor (from table) $= 151.036$

PVA (1st annuity) $= 2000 \times 151.036 = \$302\,072$

2nd Annuity

Find PVA for \$1200 paid monthly from year 16 to 25:

PVA (2nd annuity) = PVA (25 years) − PVA (15 years)

$$r = \frac{2.4}{12} = 0.2\% = 0.002$$

Total payments (25 years) $= 25 \times 12 = 300$

PVA factors (from table): 225.430 (25 years), 151.036 (15 years)

PVA (2nd annuity) $= (1200 \times 225.430) - (1200 \times 151.036)$

$= \$89\,272.80$

$\therefore$ Minimum deposit $= 302\,072 + 89\,272.80 = \$391\,344.80$

♦ Mean mark 51%.

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30. Financial Maths, STD2 F5 2015 HSC 30c

i. $N = 74, r = 0.0080$

PV(annuity) table factor $= 55.68446$

$\therefore PV$ of annuity

$= \$200 \times 55.68446$

$= \$11\ 136.892$

$= \$11\ 136.89$ (nearest cent)

♦ Mean mark 48%.

ii. Over 6 years

$N = 6 \times 12 = 72$ months

$$r = \frac{10.8}{12} = 0.9\% = 0.009$$

PV(annuity) table factor $= 52.82118$

Let $\$M =$ monthly repayment

Loan $= PV$ of annuity

$\$21\ 500 = M \times 52.82118$

$\therefore M = \$407.033\ldots$

$= \$407$ (nearest dollar)

♦♦ Mean mark 33%.

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31. Financial Maths, STD2 F5 2016 HSC 28d

Periods $= 6 \times 4 = 24$

Interest rate $= \dfrac{1}{4} \times 3 = 0.75\%$

$\Rightarrow$ Table factor $= 0.0382$

(i.e. 3.82 cents contributed per quarter $= \$1$ after 6 years)

$\therefore$ Quarterly contribution

$= 75\ 000 \times 0.0382$

$= \$2865$

♦ Mean mark 40%.

32. Financial Maths, STD2 F5 2019 HSC 42

$$\text{Annuity compounding factor } (4\% \text{ for 6 years}) = 6.633$$

$$\begin{aligned}\therefore \text{ Value after 6 years} &= 2000 \times 6.633\\ &= \$13\ 266.00\end{aligned}$$

♦♦ *Mean mark 27%.*

At the end of 7th year:

$$\begin{aligned}\text{Value} &= 13\ 266 \times 1.05 + 2000\\ &= 13\ 929.30 + 2000\\ &= \$15\ 929.30\end{aligned}$$